The Factor Zoo

因子动物园

This tutorial is the theoretical spine of Part II. Every subsequent factor tutorial (05 value, 06 momentum, 07 quality, 08 other) drills into one family; this one gives you the asset-pricing lineage, the taxonomy, the "why do factors earn premia" debate, and the sober statistical reality of the factor zoo — so that when you read a paper claiming a new anomaly with a t-stat of 2.1, you know exactly how skeptical to be.

The completion pass adds the missing bridge from theory to practice: GRS intuition, multiple-testing discipline, factor taxonomy, construction styles, factor timing/crowding, US-vs-A-share differences, and a practical checklist for judging whether a factor deserves further work.

因子动物园

本教程是第二部分的理论主干。后续每一篇因子教程（ 05 价值、 06 动量、 07 质量、 08 其他因子）都会深入研究某一个因子家族；而本教程负责搭建资产定价的理论谱系、因子分类体系、 " 因子为何能获得溢价 " 的争论，以及对 " 因子动物园 " 这一统计现实的清醒认知——这样当你读到一篇宣称发现新异象、 t 统计量为 2.1 的论文时，才能准确判断应该保持多大程度的怀疑。本次补全增加了从理论到实践的缺失桥梁： GRS 直觉、多重检验纪律、因子分类、构建方式、因子择时 / 拥挤、美股与 A 股差异，以及判断一个因子是否值得继续研究的实践清单。

Definitions

$$r_{i,t} = E[r_{i,t}] + \sum_{k=1}^{K} \beta_{i,k,t} f_{k,t} + \epsilon_{i,t}$$

then f_(k,t) is a factor return at time t, β_(i,k,t) is stock i's loading (exposure) on factor k, and ε_(i,t) is idiosyncratic noise, assumed uncorrelated across stocks and with the factors. A factor is useful if it captures common variation in returns (reduces the covariance matrix to a low-rank-plus-diagonal structure) and/or if exposure to it commands a non-zero risk premium λ_k = E[f_(k,t)].

在股票研究中，因子是指一种系统性的、可在横截面上观察到的特征（或由这种特征构建出的收益率序列），它能够解释股票之间预期收益率的差异。形式化地说，如果我们把股票 i 在时刻 t 的收益生成过程写成

那么 f_(k,t) 就是时刻 t 的因子收益率，β_(i,k,t) 是股票 i 对因子 k 的载荷（暴露度），ε_(i,t) 是特异性噪声，假设它在股票之间以及与因子之间都不相关。如果一个因子能够捕捉收益率中的共同变动（将协方差矩阵化简为 " 低秩加对角 " 结构），和 / 或对该因子的暴露能带来非零的风险溢价 λ_k = E[f_(k,t)] ，那么这个因子就是有用的。

Factor comparison

Term	Definition	Example
Characteristic 特征	A firm-level observable at time t (accounting ratio, price statistic, sentiment score) 公司层面在时刻 t 可观察到的量（会计比率、价格统计量、情绪评分）	Book-to-market ratio B/M_(i,t) 账面市值比 B/M_(i,t)
Factor 因子	A return series (usually long-short) built by sorting stocks on a characteristic, or a statistical factor from PCA/regression 依据某特征对股票排序构建出的收益率序列（通常为多空组合），或由 PCA/ 回归得到的统计因子	HML: high B/M minus low B/M portfolio return HML ：高 B/M 减低 B/M 组合收益

Factor comparison

Term	Definition	Example
Signal / Alpha 信号 / 阿尔法	A cross-sectional score used to rank stocks for portfolio construction, may or may not correspond to a known academic factor 用于给股票排序以构建组合的横截面评分，可能对应也可能不对应某个已知的学术因子	A proprietary blend of 40 characteristics, output as a single number per stock per day 由 40 个特征专有混合而成，每只股票每天输出一个数值

Factor comparison

Term	Definition	Example
Anomaly 异象	An empirical pattern in average returns that a benchmark asset pricing model (usually CAPM or FF3) fails to explain — i.e., a factor is "anomalous" relative to *some* model, not in absolute terms 某个基准资产定价模型（通常是 CAPM 或 FF3 ）无法解释的平均收益率经验模式——即因子相对于 * 某个 * 模型而言是 " 异常 " 的，而非绝对意义上的异常	Momentum was an anomaly relative to CAPM/FF3; it is not anomalous relative to Carhart's own 4-factor model (by construction) 动量相对于 CAPM/FF3 是异象；但相对于 Carhart 自己的四因子模型（按构造）就不是异象
Alpha (α) 阿尔法	The intercept in a factor-model regression — return unexplained by the chosen factors 因子模型回归中的截距项——所选因子无法解释的收益部分	α_i in r_i - r_f = α_i + β_i^T f + ε_i

Definitions

01 The crucial epistemic point: "anomaly" is model-relative, not absolute. A characteristic is an anomaly with respect to a specific benchmark model. As benchmark models expand (CAPM

02 FF3

03 Carhart4

04 FF5

05 FF6), yesterday's anomaly becomes today's risk factor, and what remains "anomalous" shrinks. This is why the number of "factors in the literature" and the number of factors that survive against an *expanding* set of controls are very different numbers — a theme central to Section 5.

Definitions

01 一个关键的认识论要点是：" 异象 " 是相对于模型而言的，而非绝对概念。 某个特征是否是异象，取决于所选定的基准模型。随着基准模型不断扩展（ CAPM

02 FF3

03 Carhart4

04 FF5

05 FF6 ），昨天的异象会变成今天的风险因子，" 仍然异常 " 的部分也随之缩小。这正是为什么 " 文献中出现过的因子数量 " 与 " 在不断扩大的控制变量集合下依然存活的因子数量 " 是两个截然不同的数字——这一主题将贯穿第 5 节的讨论。

- Risk factors 风险因子 : exposure to f_k genuinely raises the volatility of your consumption/wealth in bad states of the world (recessions, liquidity crunches, tail events). The premium $\lambda_k > 0$ is compensation demanded by rational, risk-averse investors, exactly as in CAPM/APT. Such premia should be permanent (not arbitraged away) because bearing the risk is unavoidable in equilibrium.

风险因子与错误定价因子

风险因子：对 f_k 的暴露会在坏时期（衰退、流动性紧缩、尾部事件）真实地放大你的消费 / 财富波动性。溢价 λ_k > 0 是理性、厌恶风险的投资者所要求的补偿，正如 CAPM/APT 所描述的那样。这类溢价应当是永久性的（不会被套利消除），因为在均衡状态下承担该风险是不可回避的。

Most academically documented factors are argued, by different camps, to be some mixture of both. Section 4 develops all three competing explanations (risk-based, behavioral, frictions-based) with the specific factors each explains best.

学术文献中记录的大多数因子，都被不同阵营的学者认为是这两种成分的某种混合。第 4 节将展开三种相互竞争的解释（基于风险、基于行为、基于摩擦），并说明每种解释最适合解释哪些具体因子。

CAPM

$$E[r_i]-r_f=\beta_i(E[r_m]-r_f),\beta_i=\frac{Cov(r_i,r_m)}{Var(r_m)}$$

值得简要推导一遍这个模型，因为它所依赖的假设，正是后来在实证中被证伪的那些假设。考虑 N 项风险资产，预期收益向量为 μ，协方差矩阵为 Σ，外加一个无风险利率 r_f 的无风险资产。一位均值 - 方差投资者求解：

$$max_w\ w^\top\mu-\frac{\gamma}{2}w^\top\Sigma w\ s.t.\ w^\top 1=1$$

It is worth sketching the derivation, because the assumptions it relies on are exactly the assumptions that later fail empirically. Consider N risky assets with expected return vector μ and covariance matrix Σ, plus a riskless asset with rate r_f. A mean-variance investor solves:

CAPM

$$w^* = \frac{1}{\gamma} \Sigma^{-1} (\mu - r_f 1)$$

where γ is the investor's risk-aversion coefficient and w is the vector of portfolio weights on the risky assets (with 1-w^T\mathbf 1 held in the riskless asset when the budget constraint is relaxed to allow cash). The first-order condition gives the optimal risky-asset weights as:

其中 γ 是投资者的风险厌恶系数，w 是风险资产组合权重向量（若放松预算约束允许持有现金，则 1-w^T\mathbf 1 部分配置于无风险资产）。一阶条件给出最优风险资产权重：

Every mean-variance investor, regardless of risk aversion γ, holds risky assets in the same relative proportions Σ^(-1)(μ - r_f\mathbf 1) — they differ only in how much they lever this common risky portfolio up or down against cash (the two-fund separation theorem).

CAPM

$$w_{mkt} \propto \Sigma^{-1}(\mu - r_f 1) \Longrightarrow \mu - r_f 1 \propto \Sigma w_{mkt}$$

无论风险厌恶系数 γ 为何值，每一位均值 - 方差投资者持有风险资产的相对比例都是相同的，即 Σ^(-1)(μ - r_f\mathbf 1)—— 投资者之间的差异仅在于如何用现金对这个共同的风险组合加杠杆或去杠杆（这就是两基金分离定理）。如果所有投资者拥有相同的信念（μ、Σ），且每个投资者的风险资产持仓都与同一个向量成比例，那么在均衡状态下，将所有投资者的持仓加总后，市场总风险组合必然等于这个共同的比例组合——又因为市场必须出清（每项风险资产的总需求等于其市值权重），市值加权组合本身必然就是这个共同的最优组合：

If all investors have the same beliefs (μ, Σ) and every investor's risky holdings are proportional to the same vector, then in equilibrium, when we aggregate over all investors, the aggregate risky portfolio must equal this common proportional portfolio — and since the market must clear (aggregate demand for each risky asset equals its market-cap weight), the market-capitalization-weighted portfolio itself must be that common optimal portfolio:

1. Only beta matters. Idiosyncratic risk is diversifiable and unpriced.

CAPM

只有贝塔重要。 特异性风险可以被分散化消除，因而不应被定价。

截距等于 r_f ，斜率等于市场风险溢价——一旦控制了贝塔，任何其他变量都不应再具有解释力。

Frazzini and Pedersen (2014) later formalized *why* this happens: many real-world investors (pension funds, retail investors, mutual funds under mandate) face leverage constraints — they cannot borrow freely to lever up a low-beta portfolio to their desired risk level, so instead they overweight high-beta stocks directly as a leverage substitute.

3. The intercept equals r_f and the slope equals the market risk premium — nothing else should have explanatory power once beta is controlled for.

- Flat/too-flat SML — the betting-against-beta puzzle. Black, Jensen, and Scholes (1972) sorted stocks into beta-ranked portfolios and found that realized average returns were far less sensitive to beta than CAPM predicted — the empirical line was flatter than the theoretical SML, with low-beta stocks earning more, and high-beta stocks earning less, than CAPM implied.

CAPM

This constrained-investor demand bids up high-beta stocks (lowering their expected return) and leaves low-beta stocks relatively underbought (raising their expected return relative to CAPM) — producing exactly the flat SML observed empirically, and motivating a Betting-Against-Beta (BAB) factor that goes long leverage-adjusted low-beta stocks and short leverage-adjusted high-beta stocks. This is the empirical seed of what became the low-volatility / betting-against-beta anomaly (Tutorial 08).

CAPM

证券市场线过于平坦——"贝塔博弈"之谜。 Black、 Jensen 和 Scholes（ 1972 ）按贝塔对股票分组构建组合，发现已实现的平均收益率对贝塔的敏感度远低于 CAPM 的预测——实证得到的直线比理论 SML 更平坦，低贝塔股票的收益高于 CAPM 隐含值，高贝塔股票的收益则低于 CAPM 隐含值。Frazzini 和 Pedersen（ 2014 ）后来正式论证了 * 为什么 * 会出现这种现象：现实中许多投资者（养老基金、散户、受托管约束的共同基金）面临杠杆约束——他们无法自由借贷来把低贝塔组合加杠杆到理想的风险水平，于是转而直接超配高贝塔股票，作为杠杆的替代手段。这种受约束投资者的需求推高了高贝塔股票的价格（压低其预期收益），同时使低贝塔股票相对少被购买（相对于 CAPM 抬高其预期收益）——恰好产生了实证中观察到的平坦 SML ，并催生了"贝塔博弈"（ Betting-Against-Beta, BAB ）因子：做多经杠杆调整的低贝塔股票，做空经杠杆调整的高贝塔股票。这正是后来演变为低波动率 / 贝塔博弈异象（教程 08 ）的实证源头。

- Size effect (Banz 1981): small-cap stocks earned returns systematically higher than their CAPM beta justified.

CAPM

$$r_i = E[r_i] + \sum_{k=1}^{K} \beta_{ik} f_k + \epsilon_i$$

罗尔批判（ 1977 ）：真正的 " 市场组合 "（涵盖股票、债券、房地产、人力资本在内的全部财富）是不可观测的，因此仅用股票市值加权指数作为代理变量来检验 CAPM ，在原则上是不可行的。

$$E[r_i] \approx r_f + \sum_{k=1}^{K} \beta_{ik} \lambda_k$$

其中 f_k 是均值为零的共同因子（宏观冲击，或者 * 任何 * 在众多资产间普遍存在的共同驱动因素），ε_i 是特异性的、可分散化的。 APT 并不规定这些因子究竟 * 是什么 *—— 这是一个实证问题——但它证明了，在一个规模庞大、充分分散化的市场中，无套利条件将迫使预期收益（近似地）成为因子载荷的线性函数：

APT

where λ_k is the risk premium associated with factor k. The proof sketch: construct a well-diversified portfolio with zero net investment, zero loading on every factor, and non-zero expected return — if such a portfolio existed, it would be a riskless arbitrage; no-arbitrage forces its expected return to zero, which algebraically forces the linear pricing relation above (up to a vanishing diversifiable error as $N \to \infty$).

其中 λ_k 是与因子 k 相关联的风险溢价。证明概要：构造一个净投资为零、对每个因子的载荷均为零、但预期收益非零的充分分散化组合——如果这样的组合存在，它就构成了一个无风险套利机会；无套利条件迫使其预期收益必须为零，这在代数上就迫使上述线性定价关系成立（当 $N \to \infty$ 时，可分散化误差趋于消失）。

APT

$$r_i - r_f = \alpha_i + \beta_i MKT + s_i SMB + h_i HML + \epsilon_i$$

$$HML = \frac{1}{2}(S/H + B/H) - \frac{1}{2}(S/L + B/L)$$

$$SMB = \frac{1}{3}(S/L + S/N + S/H) - \frac{1}{3}(B/L + B/N + B/H)$$

APT 对从业者的重要性是 * 方法论层面 * 的，而不仅仅是历史意义：它使得 " 寻找 K 个能对横截面定价的共同因子 " 这整套研究范式变得正当合理，而无需像 CAPM 那样先从代表性主体的效用最大化问题中推导出这些因子。下文的 Fama-French 、 Carhart 以及 q 因子模型，在精神上都属于 APT 式的实证因子模型——用少量可交易的多空组合来替代不可观测的 f_k 。

因子随后定义为：

APT's importance for practitioners is *methodological*, not just historical: it legitimizes the entire enterprise of "find K common factors that price the cross-section" without requiring you to first derive them from a representative-agent utility maximization problem, as CAPM does. Fama-French, Carhart, and the q-factor model below are all, in spirit, APT-style empirical factor models — a small number of tradeable long-short portfolios standing in for the unobserved f_k.

Fama-French 三因子模型

$$r_i - r_f = \alpha_i + \beta_i MKT + s_i SMB + h_i HML + m_i UMD + \epsilon_i$$

$$r_i - r_f = \alpha_i + \beta_i MKT + s_i SMB + h_i HML + r_i^{RMW} RMW + c_i CMA + \epsilon_i$$

采用独立的 2×3 设计（而非例如 5×5 双重分组）是刻意为之的：由于 SMB 和 HML 各自都是对 * 另一个 * 排序变量各档位取平均，这一设计使得两者在构造上就近似正交。

构建方法。 每月末，按 t-12 到 t-2 月的累计收益率对股票排序（跳过最近一个月以避免短期反转的污染）。构建市值加权的多空组合（通常为前 30% 减后 30% ，或最高十分位减最低十分位），常常在规模分组内部进行（一个类似于 SMB/HML 的 2×3 分组，例如按 Kenneth French 数据库的惯例）。与每年重新构建的 HML/SMB 不同， UMD 按月再平衡，这反映了动量信号半衰期更短、换手率更高的特点（第 7 节）。
RMW （ " 强健减弱势 " ）：经营盈利能力因子。经营盈利能力定义为 （营业收入 - 营业成本 - 销售管理费用 - 利息费用)/ 账面权益。强健（高）减弱势（低）盈利能力，市值加权，采用与 HML 相同的 " 规模 +2×3 分组 " 方法论。

Fama-French 五因子模型

$$M_t = \sum_{\tau=1}^{\infty} \frac{E\left[Y_{t+\tau} - d\,B_{t+\tau}\right]}{\left(1+r\right)^{\tau}}$$

where M_t is market equity, Y_(t+τ) is expected future earnings, and dB_(t+τ) is the expected change in book equity (investment), Fama and French show algebraically that, holding M_t (equivalently B_t/M_t) and expected profitability fixed, a higher expected rate of investment dB implies a *lower* discount rate r (since more of the earnings stream is being reinvested rather than paid out, for a fixed price today) — hence "conservative investment → higher expected return" is not a behavioral add-on but a restatement of the same present-value logic that produces the value premium.

This is why FF5's R^2 against a wide set of test portfolios rose further, and why HML's marginal contribution becomes largely redundant once RMW and CMA are included (FF5 alphas on HML-sorted portfolios shrink toward zero — the "HML is redundant" result that generated its own controversy).

Fama-French 五因子模型

$$r_i - r_f = \beta_i^{MKT} MKT + \beta_i^{ME} r^{ME} + \beta_i^{I/A} r^{I/A} + \beta_i^{ROE} r^{ROE} + \epsilon_i$$

- I/A — investment-to-assets factor: I/A = Δ Total Assets / Total Assets_(t-1), low-minus-high (conservative-minus-aggressive, conceptually parallel to CMA but with a different exact definition and triple-sort construction).

I/A—— 资产投资比因子： I/A = Δ 总资产 / 总资产 _(t-1) ，低减高（保守减激进，概念上与 CMA 平行，但具体定义和三重分组构建方式不同）。

其中 M_t 是市值， Y_(t+τ) 是预期未来盈余， dB_(t+τ) 是账面权益的预期变化（即投资）。Fama 和 French 用代数方法证明：在固定 M_t （等价地固定 B_t/M_t ）和预期盈利能力的条件下，更高的预期投资增速 dB 意味着 * 更低 * 的折现率 r （因为在今天价格固定的情况下，盈余流中更大比例被再投资而非派发出去）——因此 " 保守投资 → 更高预期收益 " 并非一个行为金融学的附加解释，而只是产生价值溢价的同一套现值逻辑的另一种表述。这正是为什么 FF5 在众多测试组合上的 R^2 进一步提高，也是为什么一旦纳入 RMW 和 CMA ， HML 的边际贡献就基本变得多余（ FF5 模型下按 HML 分组的组合阿尔法趋近于零——这一 "HML 是冗余的 " 结论本身也引发了不小的争议）。

q 因子模型

- ROE — return-on-equity factor: quarterly ROE = Income before extraordinary items / Book Equity (most-recent-quarter, exploiting the faster information diffusion of quarterly earnings), high-minus-low.

q-theory rationale (brief). In a standard investment-based model, a firm invests until the marginal cost of investment equals marginal Q (the present value of the incremental cash flows the investment generates).

Firms with high expected profitability (ROE) and low expected new investment (I/A) are, in equilibrium, firms whose discount rate (cost of capital, i.e., expected stock return) must be high enough that they choose *not* to invest more — this again derives value/profitability/investment premia from a no-arbitrage-plus-production-technology argument, structurally parallel to, but derived independently of, Fama-French's dividend-discount argument. That two very different theoretical routes (consumption-side present value vs.

q 因子模型

$$r_i - r_f = \beta^{MKT} MKT + \beta^{ME} r^{ME} + \beta^{I/A} r^{I/A} + \beta^{ROE} r^{ROE} + \beta^{EG} r^{EG} + \epsilon_i$$

production-side investment theory) arrive at "profitability up, investment down ⇒ higher expected return" is considered a meaningful cross-validation in the literature.

q 理论的直觉（简述）。 在标准的基于投资的模型中，企业会持续投资，直到投资的边际成本等于边际 Q （该投资所产生的增量现金流的现值）。预期盈利能力（ ROE ）高、预期新增投资（ I/A ）低的企业，在均衡状态下，其折现率（资本成本，即预期股票收益率）必然足够高，以至于它们 * 选择不 * 进行更多投资——这再一次从 " 无套利加生产技术 " 的论证中推导出价值 / 盈利能力 / 投资溢价，其结构与 Fama-French 的股利折现论证平行，但却是独立推导出来的。两条截然不同的理论路径（消费端的现值逻辑与生产端的投资理论）都得出 " 盈利能力越高、投资越低 ⇒ 预期收益越高 " 这一结论，这在文献中被视为一种有意义的交叉验证。

EG 由分析师一致预期的长期盈利增长预测与已实现基本面数据结合构建而成，其作者证明它显著提升了模型对动量等 q4 无法完全吸收的 " 前瞻性 " 异象的定价能力——这也是对 " 即便 q4 也留有余地 " 的明确承认。

SY 错误定价因子

$$r_i - r_f = \alpha_i + \beta_i MKT_i + s_i SMB_i + m_i MGMT + p_i PERF + \epsilon_i$$

Stambaugh and Yuan (2017) took an explicitly *behavioral* rather than risk-based or production-based approach. Instead of hand-picking one characteristic per factor, they clustered 11 well-known anomaly characteristics (associated with, e.g., accruals, net stock issuance, asset growth, momentum, gross profitability, ROA, ROE, etc.) into two correlated clusters via factor analysis of pairwise correlations of characteristic-sorted portfolio returns, then built two composite "mispricing" factors:

Stambaugh 和 Yuan（2017）采取了一种明确 * 基于行为 * 而非基于风险或基于生产的路径。他们没有为每个因子手工挑选单一特征，而是通过对特征分组组合收益率的两两相关性进行因子分析，将 11 个知名异象特征（涉及应计项目、股票净发行、资产增长、动量、毛利率、 ROA 、 ROE 等）聚类为两个相关簇，进而构建两个复合 " 错误定价 " 因子：

SY 错误定价因子

The explicit theoretical commitment here is that each cluster shares a common mispricing driver correlated with *arbitrage risk and investor sentiment* (their factors interact strongly with measures like the Baker-Wurgler sentiment index — mispricing effects are stronger following high-sentiment periods, consistent with overpricing being harder to arbitrage away than underpricing due to short-sale constraints).

This model is a direct competitor to FF5/q5 in "anomaly explaining" horse races and routinely wins on that dimension (higher fraction of anomalies whose alphas it eliminates) *precisely because it was constructed post hoc from the anomalies it is asked to explain* — a good illustration of the multiple-testing issues taken up in Section 5.

SY 错误定价因子

这里明确的理论立场是：每个簇共享一个与 * 套利风险和投资者情绪 * 相关的共同错误定价驱动因素（他们的因子与 Baker-Wurgler 情绪指数等指标有很强的交互——错误定价效应在高情绪期之后更强，这与 " 由于卖空限制，高估比低估更难被套利消除 " 的观点一致）。这一模型在 " 解释异象 " 的竞赛中是 FF5/q5 的直接竞争对手，并且通常在这一维度上胜出（能消除更高比例异象阿尔法）——但 * 恰恰是因为它是根据被要求解释的那些异象事后构建出来的 * ，这正好说明了第 5 节将讨论的多重检验问题。

模型对照表

Model	Factors	What it adds	Main limitation
CAPM	Market	One systematic risk source	Cannot explain size, value, momentum
APT	Multiple unspecified factors	Allows several priced risks	Does not identify factors
FF3	Market, size, value	Empirical cross-section fit	Misses momentum and profitability/investment
Carhart4	FF3 + momentum	Captures UMD anomaly	Momentum crash risk remains
FF5	Market, size, value, profitability, investment	Links to q-theory intuition	HML becomes partly redundant in some samples

模型对照表

Model	Factors	What it adds	Main limitation
HXZ q/q5	Market, size, investment, ROE, expected growth	Investment-based economics	More complex and still sample-dependent
SY mispricing	Management, performance composites	Behavioral anomaly clustering	Requires careful multiple-testing control

GRS

$$r_{p,t} - r_{f,t} = \alpha_p + \beta_p^\top f_t + \epsilon_{p,t}$$

截面检验则问特征或 beta 是否解释股票或组合平均收益差异。二者相关但不相同：某个特征可以预测收益，即使它不是干净的协方差风险暴露；某个风险暴露也可能重要，即使它的特征代理很噪。

Cross-sectional tests ask whether characteristics or betas explain differences in average returns across stocks or portfolios. The two are related but not identical: a characteristic can forecast returns even if it is not a clean covariance risk exposure, and a risk exposure can matter even if its characteristic proxy is noisy.

GRS

The Gibbons-Ross-Shanken test asks whether a set of test-asset alphas is jointly zero under a factor model. In practice, it is a useful discipline because it prevents "one pretty alpha" from hiding the fact that the model fails broadly across the test assets. But it is sensitive to test-asset choice and assumes stable covariance estimates, so it should be one diagnostic, not the final word.

因子为何能获得溢价？

1. Risk compensation: investors require a premium for holding assets that perform badly in bad states.
2. Behavioral mispricing: investors overreact, underreact, extrapolate, or prefer lottery-like payoffs.
3. Market frictions: limits to arbitrage, shorting costs, funding constraints, taxes, mandates, and benchmark pressure prevent immediate correction.

No serious practitioner needs a factor to belong purely to one camp. What matters is whether the story predicts when the factor should fail. Value can fail when distress risk is realized; momentum can crash at sharp rebounds; low volatility can suffer when leverage constraints relax; size can fail when the microcap premium becomes crowded or untradeable.

严肃从业者不需要一个因子完全属于某一阵营。真正重要的是，这个故事能否预测因子何时失效。价值会在困境风险兑现时失效；动量会在剧烈反弹时崩溃；低波动会在杠杆约束放松时受损；规模会在微盘溢价拥挤或不可交易时失效。

因子动物园

The factor zoo problem is simple: hundreds of published anomalies cannot all be independent sources of expected return. Some are aliases for the same latent exposure, some are subsumed by known factors, some are too expensive to trade, and some are false discoveries produced by multiple testing.

因子动物园问题很简单：数百个已发表异象不可能都是独立预期收益来源。其中一些是同一潜在暴露的不同名称，一些会被已知因子解释，一些交易成本过高，还有一些是多重检验产生的虚假发现。

因此，新因子的最低标准高于 "t 统计量大于 2"。它应有经济逻辑、时点正确构建、跨时期和股票池稳健、与已有因子低相关、扣除成本后可实施，并有考虑多重检验的验证记录。

Factor comparison

Factor	Risk story	Mispricing story	Data-mining concern
Value	Distress and duration risk	Extrapolation and overreaction	Many metric choices
Momentum	Crash exposure and optionality	Underreaction and slow diffusion	Parameter grids
Quality	Defensive cash flows	Underappreciated persistence	Definition proliferation
Investment	q-theory investment risk	Overinvestment and empire building	Accounting variants
Low volatility	Leverage constraints	Lottery demand for high beta	Hidden sector exposure
Size	Illiquidity/funding risk	Neglected firms	Microcap and cost bias

Factor comparison

Factor	Risk story	Mispricing story	Data-mining concern
Liquidity	Compensation for trading frictions	Attention and segmentation	Measured spread quality

因子分类表

Family	Representative signals	Typical controls
Value	B/P, E/P, EBIT/EV, CFO/P	Industry, profitability, leverage
Momentum	12-1 return, residual momentum, earnings momentum	Reversal, volatility, crash risk
Quality	GP/A, ROE, accruals, F-score, QMJ	Sector, leverage, size
Investment	asset growth, capex growth, net issuance	Profitability and financing
Low risk	beta, idiosyncratic vol, downside vol	Sector and leverage
Liquidity	Amihud, turnover, spread, ADV	Size and price
Sentiment	analyst revisions, short interest, news tone	Coverage and attention

因子分类表

Family	Representative signals	Typical controls
Alternative data	web traffic, app usage, supply chain, text	Coverage bias and timestamping

因子组合构建方式

The same factor can be expressed several ways: long-short decile portfolios, rank-weighted dollar-neutral portfolios, long-only tilts versus a benchmark, residualized signals inside an optimizer, or factor-mimicking portfolios for risk models. These are not interchangeable. A long-short academic factor tests a premium; a long-only index enhancement portfolio tests whether the signal improves benchmark-relative stock selection under constraints.

同一个因子可以有多种表达方式：多空十分位组合、排名加权美元中性组合、相对基准的多头倾斜、优化器中的残差信号，或风险模型中的因子模拟组合。它们不可互换。学术多空因子检验溢价；多头指数增强组合检验信号能否在约束下提升相对基准选股。

因子择时与因子拥挤

Factor timing is tempting and difficult. Valuation spreads, recent factor returns, macro regimes, volatility, and crowding indicators can all contain information, but they also add another layer of multiple testing. The safer default is not aggressive factor timing; it is diversification across factors with drawdown-aware risk budgeting.

因子择时很诱人，也很困难。估值利差、近期因子收益、宏观状态、波动率和拥挤度指标都可能包含信息，但也会增加一层多重检验。更安全的默认做法不是激进择时，而是跨因子分散，并进行回撤敏感的风险预算。

拥挤不同于估值差。拥挤因子可以持续有效，直到流动性消失，然后剧烈反转。在数据允许时，应监控共同持仓、 ETF/ 指数资金流、空头兴趣、借券成本、换手率跳升，以及因子管理人之间的相关性。

美股与 A 股的因子表现对比

US and A-share factor behavior differs because market structure, investor base, shorting constraints, disclosure quality, and regulatory regimes differ. A factor imported from one market should be treated as a hypothesis, not a parameter-free truth.

例如：美股动量是经典稳健因子，而 A 股动量更依赖期限和涨跌停事件；A 股规模因子受壳价值和量化拥挤塑造；A 股质量筛选必须更关注补贴、关联方活动和盈余管理动机。

算例

$$HML=\frac{1}{2}(SmallHigh+BigHigh)-\frac{1}{2}(SmallLow+BigLow)$$

独立分组的目的，是避免某个因子仅仅成为规模代理。同一模板可用于盈利能力、投资因子和许多自定义因子。

从因子到策略

A factor is not yet a strategy. A strategy needs universe rules, forecasts, portfolio construction, execution assumptions, risk limits, capacity estimates, monitoring, and governance. Many factors with beautiful long-short academic returns are unattractive after borrow costs, turnover, benchmark constraints, and capacity limits.

因子还不是策略。策略需要股票池规则、预测、组合构建、执行假设、风险限制、容量估计、监控和治理。许多学术多空收益漂亮的因子，在加入借券成本、换手、基准约束和容量限制后并不吸引。

从因子到策略

01 The workflow is: economic rationale

02 clean signal

03 single-factor test

04 incrementality test

05 backtest

06 red team

07 constrained portfolio

08 paper trade

09 small live allocation.

01 流程应为：经济逻辑

02 干净信号

03 单因子检验

04 增量性检验

05 回测

06 红队检验

07 约束组合

08 模拟盘

09 小规模实盘。

实践清单：什么才是好因子

- It is known at the time it is used.
- It has a reason to exist before seeing the backtest.
- It survives neutralization to obvious exposures.
- Its IC, portfolio sorts, and regression tests agree directionally.
- It works across reasonable horizons and universes.
- It is not just a renamed existing factor.
- It survives costs and turnover.
- It has a failure story and monitoring plan.

实践清单：什么才是好因子

- 使用时确实已知。
- 在看到回测前就有存在理由。
- 对明显暴露中性化后仍存在。
- IC 、组合分层和回归检验方向一致。
- 在合理期限和股票池中稳健。
- 不是已有因子的改名。
- 能扛住成本和换手。
- 有失效故事和监控方案。

Summary

The factor zoo is both a library and a warning. It gives researchers a vocabulary for systematic return patterns, but it also shows how easily finite noisy data can produce beautiful false discoveries. Good factor research therefore combines theory, data discipline, conservative testing, cost realism, and humility about regimes.

因子动物园既是资料库，也是警告。它为系统性收益模式提供语言，但也说明有限噪声数据多么容易产生漂亮的虚假发现。因此，好的因子研究必须结合理论、数据纪律、保守检验、成本现实，以及对市场状态的谦逊。

Factor comparison

Model 模型	Year 年份	Factors 因子	#	Theoretical grounding 理论基础
CAPM	1964	MKT	1	Mean-variance equilibrium 均值 - 方差均衡
APT	1976	Unspecified f_1,…,f_K 未指定 f_1,…,f_K	K	No-arbitrage, linear factor structure 无套利、线性因子结构
Fama-French 3	1993	MKT, SMB, HML	3	Empirical + risk-based rationalization 实证 + 基于风险的理论化
Carhart 4	1997	MKT, SMB, HML, UMD	4	Empirical (momentum add-on) 实证（附加动量）

Factor comparison

Model 模型	Year 年份	Factors 因子	#	Theoretical grounding 理论基础
Fama-French 5	2015	MKT, SMB, HML, RMW, CMA	5	Dividend-discount-model decomposition 股利折现模型分解
q-factor (HXZ)	2015	MKT, ME, I/A, ROE	4	Investment-based (q-theory) 基于投资（ q 理论 ）
q5	2021	+ EG	5	q-theory + forward-looking expected growth q 理论 + 前瞻性预期增长
Stambaugh-Yuan	2017	MKT, SMB, MGMT, PERF	4	Behavioral mispricing clusters 行为金融错误定价簇